Wasteland Pantry: Concept Note

An AI/ML system for predicting and reducing food waste in kitchens by Ashish Jakhmola

In one sentence

Wasteland Pantry predicts how many kilograms of food a planned meal will waste, so kitchens can cook just enough.

What it is

Wasteland Pantry is a machine learning project. It learns from past meal records and uses the plan for an upcoming meal to estimate how much food will go to waste.

What goes in: meal time (breakfast, lunch, dinner), menu category, weekday, event type (such as exam period or festival), holiday status, expected attendance, meals planned, menu popularity, portion size, weather forecast, and waste from similar earlier meals.

What comes out: a predicted food waste figure in kilograms.

It is built with Python, pandas and scikit-learn. Two models (Linear Regression and Random Forest) are compared against simple baseline guesses, and they are tested on *future* dates, the way a real kitchen would use them.

The name comes from the idea of surviving by managing supplies carefully, a theme borrowed from post-apocalyptic survival games. The project is independent and not connected to any game or its developers.

Why it was built

The problem. Most kitchens plan quantities by experience and guesswork. But how many people turn up changes with exams, events, holidays, the menu and even the weather. When kitchens cook too much, the extra food, money, ingredients and effort are wasted.

The gap. Waste is usually written down after it has already happened, in a logbook or spreadsheet. That tells a manager what went wrong yesterday, but gives no help in deciding how much to cook tomorrow.

The idea. A forecast changes the order of things. If a manager can see the likely waste *before* cooking, they can adjust the quantity while it still matters.

The bigger goal. The project supports UN Sustainable Development Goal 12, Responsible Consumption and Production, which includes cutting food waste.

Who it is for

Main users: managers and committees running college canteens and hostel messes.

Other users: office and corporate cafeterias, community kitchens and NGOs.

Early adopters: hostel mess committees and campuses that already keep daily meal records, since the model needs history to learn from.

Who else benefits: the people eating there (less food shortage and better planning), the institution (lower food cost), and the environment (less wasted food).

The idea in use

The day before, a mess manager enters the plan for tomorrow's lunch: the menu type, the expected attendance, whether it is an exam week, the portion size, and the forecast. The model returns a predicted waste figure in kilograms. If that figure looks high, the manager can reduce the planned meals or change the portion plan. Over time, the kitchen's own records make the predictions better.

What has been built so far

A working prototype notebook that covers the full process: data checks, exploratory charts, choosing inputs that avoid data leakage, a date-based train/test split, two baselines, two models, evaluation (MAE, RMSE and R^2), error analysis and feature importance, plus a saved model that can predict for a new meal.

On the 362 test meals (late May to December 2025), both models beat both baselines. The average error was about 4.5 kg per meal, compared with about 7.1 kg for guessing the overall average and about 5.9 kg for guessing the average per meal time. Linear Regression and Random Forest performed almost the same, so the simpler model works just as well on this data.

Limits to be honest about

- **The data is synthetic.** The 1,800 meal records were generated, not collected from a real kitchen. The results show that the method works as a prototype. They do **not** prove that real waste would fall.
- **Real kitchens are messier.** Real accuracy would likely be lower and would need to be tested.
- **One input is an assumption.** The "prior similar wasted meal" column is assumed to use only earlier meals. A real system must make sure of this.
- **Patterns are not causes.** The model shows what is linked to waste, not why waste happens.

What comes next

1. Run a small pilot with one real hostel mess or canteen to collect genuine meal records.
2. Retrain and test the model on that real data.
3. Build a simple planner screen (for example with Streamlit) so staff can enter a plan and see the prediction.
4. Measure waste before and after to see whether the forecast actually helps.